Name of Student : Raghav Mahesh Sarda	**PRN:** 25070122***235***
Subject: Programming with Java	**Semester: III**
ASSIGNMENT No : 18	**Year: AY 2026-27**
TITLE : Java GUI Button Event Handling	

Problem Statement

Write a Java GUI program to demonstrate button click event handling.

Learning Objectives

To implement button click events using Java event handling mechanisms.

Program:

GUICalc.java

```
// Assignment 18 - Exercise 1: GUI Calculator using Java Swing
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
public class GUICalc implements ActionListener
{
   JFrame frame; // Top-level window frame
   JTextField num1txt, num2txt; // Input text fields
   JLabel resultlabel; // Label to display result
   JButton addbutton, subbutton; // Operation buttons

   GUICalc() // Constructor to build calculator GUI
   {
       frame = new JFrame("Simple Calculator"); // Creating window frame
       frame.setSize(350, 220); // Setting frame dimensions
       frame.setLayout(new GridLayout(4, 2, 10, 10)); // Setting grid
layout with spacing
       frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE); // Closing
behavior

       JLabel lbl1 = new JLabel("First Number: "); // Label
       num1txt = new JTextField(); // Text field
```

```
        JLabel lbl2 = new JLabel("Second Number: "); //
        num2txt = new JTextField();

        addbutton = new JButton("Add (+)"); // Addition button
        subbutton = new JButton("Subtract (-)"); // Subtraction button

        resultlabel = new JLabel("Result: "); // Label for showing output
        resultlabel.setFont(new Font("Arial", Font.BOLD, 14));

        addbutton.addActionListener(this); // Registering addition button
listener
        subbutton.addActionListener(this); // Registering subtraction
button listener

        frame.add(lbl1); // Adding label
        frame.add(num1txt); // Adding textfield
        frame.add(lbl2);
        frame.add(num2txt);
        frame.add(addbutton);
        frame.add(subbutton);
        frame.add(resultlabel);

        frame.setVisible(true); // Making window visible
    }

    public void actionPerformed(ActionEvent e) // Event handler method
    {
        try // Handling parsing errors
        {
            double n1 = Double.parseDouble(num1txt.getText());
            double n2 = Double.parseDouble(num2txt.getText());
            double res = 0; // Variable to store answer

            if(e.getSource() == addbutton) // Addition condition
            {
                res = n1 + n2;
                resultlabel.setText("Result: " + res);
            }
            else if(e.getSource() == subbutton) // Subtraction condition
            {
                res = n1 - n2;
                resultlabel.setText("Result: " + res);
            }
        }
```

```
        catch(NumberFormatException ex) // Catching invalid inputs
        {
            resultlabel.setText("Error: Enter valid numbers!");
        }
    }

    public static void main(String args[]) // Main Method
    {
        new GUICalc();
    }
}
```

BankBalanceGUICalc.java

```
// Assignment 18 - Exercise 2: Bank Balance Calculator GUI Application
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class BankBalanceGUICalc implements ActionListener
{
    JFrame frame; // Top-level window frame
    JTextField initbaltxt, transacamttxt; // Input text fields
    JLabel updatedballabel; // Label to display updated balance
    JButton depositbutton, withdrawbutton; // Action buttons
    double currentBal = 0.0; // Variable to track balance
    boolean initflag = false; // Flag to check initial balance set

    BankBalanceGUICalc() // Constructor to build GUI
    {
        frame = new JFrame("Bank Balance Calculator"); // Creating window
frame
        frame.setSize(420, 260); // Setting frame dimensions
        frame.setLayout(new GridLayout(5, 2, 10, 10)); // Setting grid
layout with padding
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE); // Closing
behavior

        JLabel initlabel = new JLabel("Initial Balance: "); // Label for
initial balance
        initbaltxt = new JTextField(); // Text field for initial balance

        JLabel transacamtlabel = new JLabel("Transaction Amount: "); //
Label for transaction amount
```

```
        transacamttxt = new JTextField(); // Text field for transaction
input

        depositbutton = new JButton("Deposit (+)");
        withdrawbutton = new JButton("Withdraw (-)");

        updatedballabel = new JLabel("Updated Balance: $0.00"); // Output
label
        updatedballabel.setFont(new Font("Arial", Font.BOLD, 13));

        depositbutton.addActionListener(this); // Registering deposit
listener
        withdrawbutton.addActionListener(this); // Registering withdraw
listener

        frame.add(initlabel);
        frame.add(initbaltxt);
        frame.add(transacamtlabel);
        frame.add(transacamttxt);
        frame.add(depositbutton);
        frame.add(withdrawbutton);
        frame.add(updatedballabel);

        frame.setVisible(true);
    }

    public void actionPerformed(ActionEvent e) // Event handler method
    {
        try
        {
            if(!initflag) // Initializing starting balance on first
action
            {
                currentBal = Double.parseDouble(initbaltxt.getText());
                initflag = true; // Setting initialized flag to true
                initbaltxt.setEditable(false);
            }

            double amt = Double.parseDouble(transacamttxt.getText());

            if(amt <= 0) // Validating positive transaction amount
            {
                JOptionPane.showMessageDialog(frame, "Please enter an
amount greater than 0.");
```

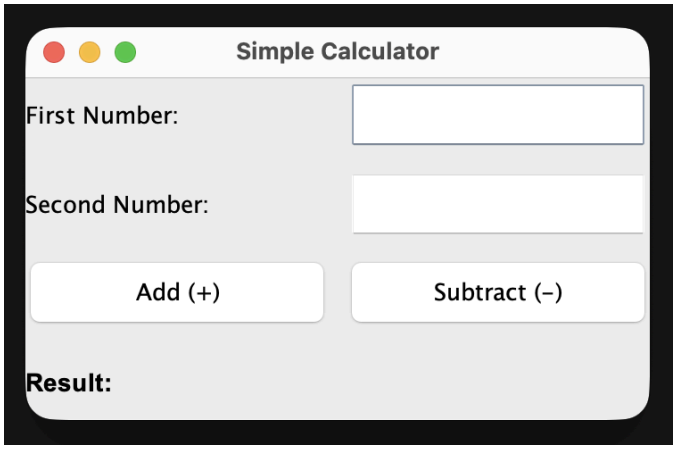
```
                return;
            }

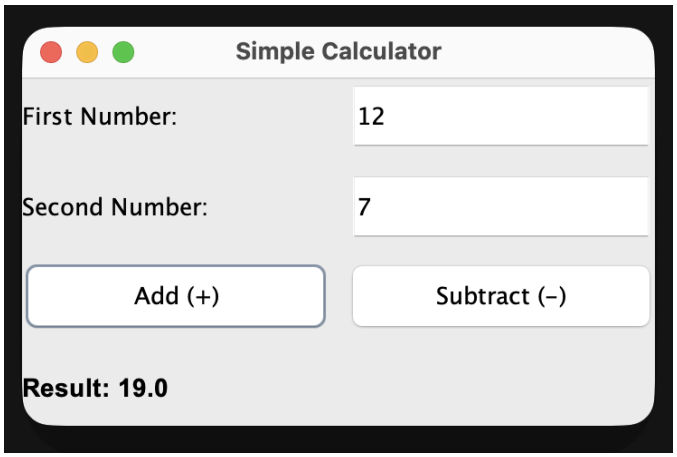
            if(e.getSource() == depositbutton) // Deposit condition
            {
                currentBal += amt; // Adding amount to balance
                updatedballabel.setText("Updated Balance: $" +
currentBal);
                transacamttxt.setText(""); // Clearing transaction field
            }
            else if(e.getSource() == withdrawbutton) // Withdraw
condition
            {
                if(amt > currentBal) // Insufficient funds condition
                {
                    JOptionPane.showMessageDialog(frame, "Insufficient
Balance for this withdrawal!");
                }
                else // Valid withdrawal
                {
                    currentBal -= amt; // Subtracting amount from balance
                    updatedballabel.setText("Updated Balance: $" +
currentBal);
                    transacamttxt.setText(""); // Clearing transaction
field
                }
            }
        }
        catch(NumberFormatException ex) // Catching invalid inputs
        {
            JOptionPane.showMessageDialog(frame, "Invalid numeric input
entered!");
        }
    }

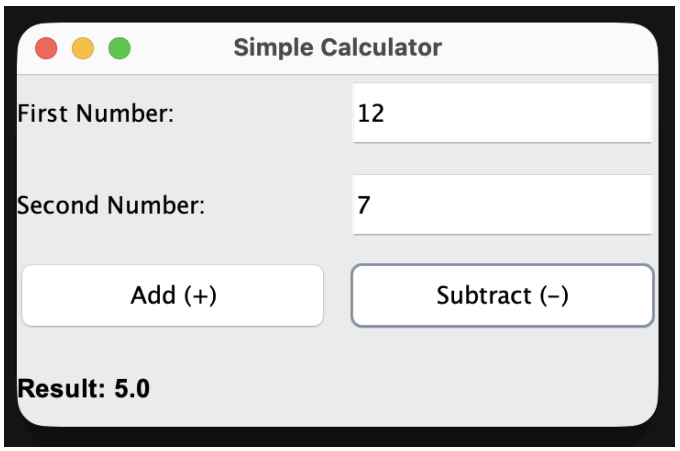
    public static void main(String args[]) // Main Method
    {
        new BankBalanceGUICalc();
    }
}
```

Output:

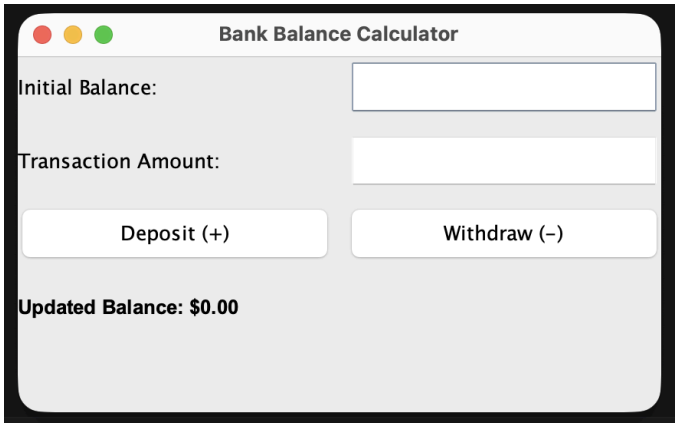
GUICalc.java

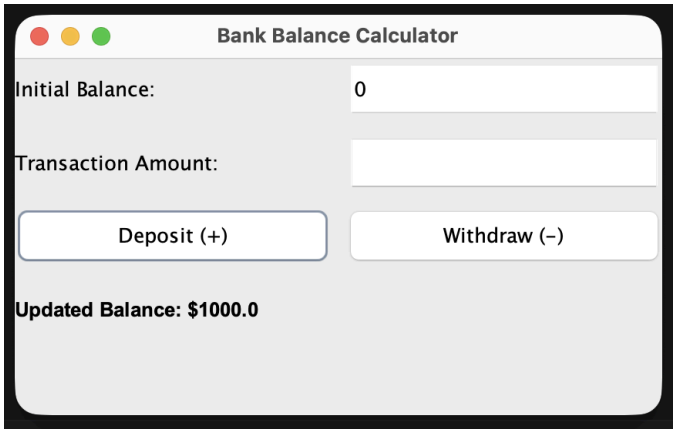


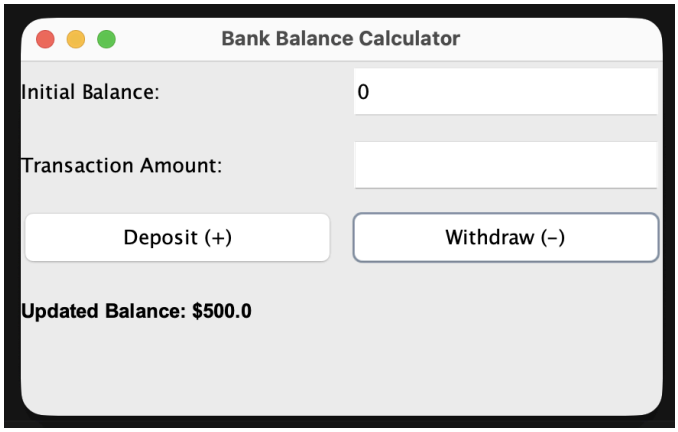




BankBalanceGUICalc.java







Conclusion:

In this assignment, the concepts of graphical user interface design and button click event handling using the Java Swing framework and the ActionListener interface were successfully implemented and verified. The GUI Calculator demonstrated basic arithmetic operations, while the Bank Balance Calculator effectively managed dynamic financial state changes through interactive deposit and withdrawal triggers, preventing invalid transactions and maintaining updated record displays.